SE LAB 03

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Scenario: Scenario - 30: Airport Lost Luggage Claim & Tracking Portal

Lab: Lab 3 – Component Modelling & Architectural Pattern Selection

Architecture Selection

We chose a **Microservices Architecture** for the Airport Lost Luggage Claim & Tracking Portal.

Architectural Choice & Justification

Microservices architecture decomposes the application into independently deployable, loosely coupled services organised around distinct business capabilities: Claim Management, Baggage Tracking, Scan Log Cross-Referencing, Notification Dispatch, and Compensation Workflows.

Reasons

1. **Independent Scalability for High-Read vs. High-Write Workloads:** Passenger anxiety creates high-volume read traffic on the tracking timeline (NFR-001 requires ≤ 200 ms latency under 1,000 concurrent requests), while baggage claims and agent location updates are write-intensive operations. Microservices allow the **Baggage Tracking Component** to be scaled independently horizontally with dedicated read-replicas, completely insulating it from resource-heavy operations like scanning cross-reference batch jobs or compensation queries.
2. **Fault Isolation Across Asynchronous Workflows:** The system relies on third-party integrations, such as external SMS/Email carrier gateways (FR-004) and airline reservation systems (FR-001). Under a microservices architecture, network latency or downtime from the **Notification Service Component** or the **Compensation Engine Component** does not degrade the core operational availability of the passenger claim intake or agent tracking dashboard.

Security Advantage

Interface Identifier	Protocol / Technology	Providing Component (Ball)	Requiring Component (Socket)	Interaction Description
IClaimService	REST over HTTPS	Claim Management Component	Passenger Web/Mobile Portal	Accepts lost luggage reports with PNR and Tag Number, returning a unique Claim ID (FR-001).
ITrackingQuery	REST / JSON	Baggage Tracking Component	Passenger Web/Mobile Portal	Delivers multi-leg scan timeline history under 200 ms latency (NFR-001).
IStatusUpdate	HTTPS / JSON	Baggage Tracking Component	Baggage Agent Console	Records manual physical location and state changes by ground agents (FR-003).
IScanMatching	Internal gRPC	Scan Cross-Reference Component	Claim Management Component	Cross-references new claims against unrouted terminal scan logs (FR-002).

Interface Identifier	Protocol / Technology	Providing Component (Ball)	Requiring Component (Socket)	Interaction Description
INotificationDispatch	AMQP / Message Queue	Notification Service Component	Claim, Tracking, & Compensation Components	Asynchronously queues alerts dispatched to passengers within 1 minute of changes (FR-004).
ISecureDataAccess	TLS / Encrypted DAO	Data Access & Encryption Component	All Core Business Services	Mediates database operations while transparently encrypting passenger PII with AES-256 (NFR-002).
IPNRValidation	SOAP / HTTPS API	External PNR System	Claim Management Component	Validates booking records and baggage tag integrity during claim submission (FR-001).
IGatewayDispatch	REST Webhook	External Carrier Gateway	Notification Service Component	Forwards SMS and email

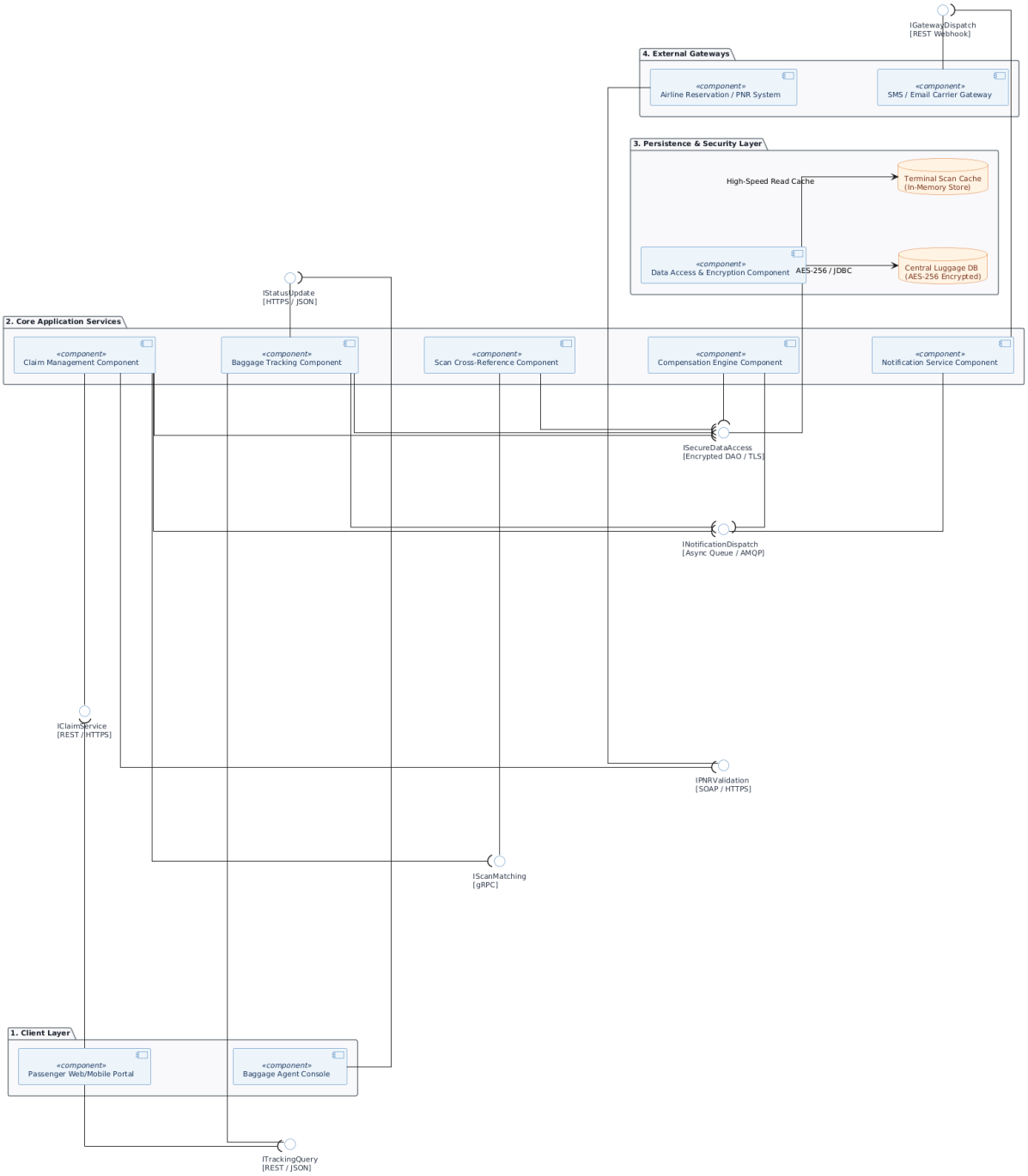
Interface Identifier	Protocol / Technology	Providing Component (Ball)	Requiring Component (Socket)	Interaction Description
				payloads out to cellular and email networks (FR-004).

The microservices pattern establishes an **isolated security boundary and data-access abstraction layer** to safeguard Passenger Personally Identifiable Information (PII), as mandated by NFR-002. Instead of exposing the database schema to all modules, only the dedicated **Data Access & Encryption Component** handles direct access to the database using AES-256 field-level encryption for passenger phone numbers and home delivery addresses. This strict isolation minimises the blast radius of potential compromises and ensures compliance with global privacy regulations like GDPR.

Performance Benefit

By isolating the **Baggage Tracking Component** into a dedicated microservice, the system can decouple timeline queries from complex transactional writes. The tracking service uses an in-memory caching layer for unrouted terminal scan logs and indexed multi-leg flight scan history. This architecture satisfies NFR-001 by guaranteeing 95th-percentile response latencies well under 200 ms for up to 1,000 concurrent requests without contending for locks against claim-creation transactions.

Component UML diagram:


IGatewayDispatch
[REST Webhook]
4. External Gateways
«component»
Airline Reservation / PNR System
«component»
SMS / Email Carrier Gateway
3. Persistence & Security Layer
High-Speed Read Cache
Terminal Scan Cache
(In-Memory Store)
«component»
Data Access & Encryption Component
AES-256 / JDBC
Central Luggage DB
(AES-256 Encrypted)
IStatusUpdate
[HTTPS / JSON]
2. Core Application Services
«component»
Claim Management Component
«component»
Baggage Tracking Component
«component»
Scan Cross-Reference Component
«component»
Compensation Engine Component
«component»
Notification Service Component
ISecureDataAccess
[Encrypted DAO / TLS]
INotificationDispatch
[Async Queue / AMQP]
IClaimService
[REST / HTTPS]
IPNRValidation
[SOAP / HTTPS]
IScanMatching
[gRPC]
1. Client Layer
«component»
Passenger Web/Mobile Portal
«component»
Baggage Agent Console
ITrackingQuery
[REST / JSON]